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ASSIGNMENT NO. 1

AIM: To Conduct a thorough security assessment on a client's network by performing host discovery, comprehensive port scanning, OS discovery, and generating a detailed report on vulnerabilities.

TARGET: www.certifiedhacker.com (IP: 162.241.216.11)

TOOL USED: ping, nmap, Wireshark

TASK:

1. Host Discovery Using the 'ping' Command:

Command: ping www.certifiedhacker.com

```
L$ ping www.certifiedhacker.com
PING www.certifiedhacker.com (162.241.216.11) 56(84) bytes of data.
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=1 ttl=51 time=294 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=2 ttl=51 time=296 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=3 ttl=51 time=277 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=4 ttl=51 time=278 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=5 ttl=51 time=295 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=6 ttl=51 time=276 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=7 ttl=51 time=284 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=8 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=9 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=10 ttl=51 time=277 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=11 ttl=51 time=286 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=12 ttl=51 time=276 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=13 ttl=51 time=289 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=14 ttl=51 time=279 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=15 ttl=51 time=278 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=16 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=17 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=18 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=19 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=20 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=21 ttl=51 time=275 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=22 ttl=51 time=277 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=23 ttl=51 time=281 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=24 ttl=51 time=285 ms
64 bytes from box5331.bluehost.com (162.241.216.11): icmp_seq=25 ttl=51 time=278 ms
^C
— www.certifiedhacker.com ping statistics —
25 packets transmitted, 25 received, 0% packet loss, time 24207ms
rtt min/avg/max/mdev = 274.756/280.282/296.446/6.688 ms
```

Key Data Extracted:

- **Target IP Address:** 162.241.216.11
- **Hostname Resolved:** box5331.bluehost.com

- **ICMP Response:** Indicates that the host is reachable.
- **Round-Trip Time (RTT):**
 1. **Minimum RTT:** 274.756 ms
 2. **Average RTT:** 280.282 ms
 3. **Maximum RTT:** 296.446 ms
 4. **Packet Loss:** 0% -No packets were lost.
- **TTL (Time to Live): 51** - Indicates the number of hops the packet has travelled before reaching the target. A TTL of 51 is commonly associated with Linux-based systems.

2. Comprehensive Port Scan:

Command: `nmap -sS -T3 -p 1-1000 -sV --version-light -O --open 162.241.216.11`

- **-sS:** SYN scan - sends SYN packets without completing the TCP handshake (stealthy and non-intrusive)
- **-T3:** Normal timing - avoids flooding the network or triggering rate-based detection
- **-p 1-1000:** Scans the top 1000 well-known ports
- **-sV:** Service versions
- **--version-light:** Detects service names safely
- **-O:** Shows Operating system
- **--open:** Shows only open ports

```
(kali㉿kali)-[~]
└─$ nmap -sS -T3 -p 1-1000 -sV --version-light -O --open 162.241.216.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 08:28 EDT
Nmap scan report for 11.216.241.162.in-addr.arpa (162.241.216.11)
Host is up (0.28s latency).
Not shown: 982 closed tcp ports (reset), 5 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          Pure-FTPd
22/tcp    open  ssh          OpenSSH 7.4 (protocol 2.0)
25/tcp    open  tcpwrapped
26/tcp    open  tcpwrapped
53/tcp    open  domain       ISC BIND 9.11.4-P2 (RedHat Enterprise Linux 7)
80/tcp    open  http         Apache httpd
110/tcp   open  pop3         Dovecot pop3d
143/tcp   open  imap         Dovecot imapd
443/tcp   open  ssl/http     Apache httpd
465/tcp   open  ssl/smtps?
587/tcp   open  smtp         Exim smtpd 4.98.1
993/tcp   open  ssl/imap     Dovecot imapd
995/tcp   open  ssl/pop3     Dovecot pop3d
Aggressive OS guesses: Android 7.1.2 (Linux 3.10) (96%), Efficient Networks SpeedStream 410
0 ADSL router (94%), Linux 2.6.28 (93%), QNAP QTS 4.2.0 (Linux 3.16) (93%), Cisco Unified C
ommunications Manager VoIP adapter (92%), Cymphonix EX550 firewall (92%), D-Link DSL-2890AL
ADSL router (92%), Draytek Vigor 2960 VPN firewall (92%), IPCop firewall 1.4.16 (Linux 2.4
.34) (92%), IPCop firewall 1.4.21 (Linux 2.4.36) (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 13 hops
Service Info: Host: box5331.bluehost.com; OS: Linux; CPE: cpe:/o:redhat:enterprise_linux:7

OS and Service detection performed. Please report any incorrect results at https://nmap.org
/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.25 seconds
```

Key Data Extracted:

- **Closed Ports:** 982 ports are closed nmap scanned it and marked as closed (rejected with RST) or filtered (no response, likely blocked by firewall).
- **Filtered Ports:** Some closed ports may be reported as filtered due to --defeat-rst-rate-limit. This indicates that some ports might be blocked by firewalls or rate-limiting mechanisms
- **Open Ports:**

Port	State	Service	Version	Notes
21/tcp	Open	FTP	Pure-FTPd	File transfer service
22/tcp	Open	SSH	OpenSSH 7.4	Remote access
25/tcp	Open	SMTP	tcpwrapped	Email service
53/tcp	Open	DNS	ISC BIND 9.11.4-P2	DNS server
80/tcp	Open	HTTP	Apache httpd	Web server
110/tcp	Open	POP3	Dovecot pop3d	Email retrieval
143/tcp	Open	IMAP	Dovecot imapd	Email retrieval
443/tcp	Open	HTTPS	Apache httpd	Secure web server
465/tcp	Open	SMTPS	ssl/http?	Secure email submission
587/tcp	Open	SMTP	Exim smtpd 4.98.1	Email submission
993/tcp	Open	IMAPS	Dovecot imapd	Secure email retrieval
995/tcp	Open	POP3S	Dovecot pop3d	Secure email retrieval

- **Web Services:**
 1. HTTP (80) and HTTPS (443) are running Apache httpd.
 2. SMTPS (465) and IMAPS (993) suggest secure email services.
- **Email Services:**
SMTP (25, 587), POP3 (110, 995), and IMAP (143, 993) indicate that this server supports multiple email protocols.
- **File Transfer:**
FTP (21) is open, which poses risks if not properly secured.
- **Remote Access:**
SSH (22) is open, allowing remote access via secure shell.
- **DNS:**
DNS (53) is running ISC BIND 9.11.4-P2, commonly used in enterprise environments.
- **Network Distance:** 13 hops mean there are 13 routers in between host and target in the network.
- **Explain the reasons for using non-intrusive methods.**

Non-intrusive methods to check the network without causing any harm. These methods do not crash services or slow down the system. They also help avoid setting off alarms in firewalls or security systems. This approach allows safe and responsible data collection without causing harm.

3. OS Discovery and Ethical/Legal Considerations:

Command: `nmap -O --osscan-guess 162.241.216.11`

- **-O:** Operating System Detection
- **--osscan-guess:** Makes the best guess if exact system is not found

```
(kali@kali)-[~]
$ nmap -O --osscan-guess 162.241.216.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 10:02 EDT
Nmap scan report for box5331.bluehost.com (162.241.216.11)
Host is up (0.28s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    open  smtp
26/tcp    open  rsftp
53/tcp    open  domain
80/tcp    open  http
110/tcp   open  pop3
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
143/tcp   open  imap
443/tcp   open  https
445/tcp   filtered microsoft-ds
465/tcp   open  smtps
587/tcp   open  submission
993/tcp   open  imaps
995/tcp   open  pop3s
1022/tcp  filtered exp2
1023/tcp  filtered netvenuechat
1026/tcp  filtered LSA-or-nterm
2222/tcp  open  EthernetIP-1
3306/tcp  open  mysql
5432/tcp  open  postgresql
9898/tcp  filtered monkeycom
Aggressive OS guesses: Android 7.1.2 (Linux 3.10) (96%), Efficient Networks SpeedStream 4100 ADSL router (94%), Linux 2.6.28 (93%), QNAP QTS 4.2.0 (Linux 3.16) (92%), Cisco Unified Communications Manager VoIP adapter (91%), Cymphonix EX550 firewall (91%), D-Link DSL-2890AL ADSL router (91%), Draytek Vigor 2960 VPN firewall (91%), IPCop firewall 1.4.16 (Linux 2.4.34) (91%), IPCop firewall 1.4.21 (Linux 2.4.36) (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 13 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.80 seconds
```

Key Data Extracted:

- **Aggressive OS guesses:** Android 7.1.2 (Linux 3.10) (96%)
- **Efficient Networks:** SpeedStream 4100 ADSL router (94%)
- Linux 2.6.28 (93%)
- QNAP QTS 4.2.0 (Linux 3.16) (92%)
- Cisco Unified Communications Manager VoIP adapter (91%)
- Cymphonix EX550 firewall (91%)
- D-Link DSL-2890AL ADSL router (91%)

- Draytek Vigor 2960 VPN firewall (91%)
- IPCop firewall 1.4.16 (Linux 2.4.34) (91%)
- IPCop firewall 1.4.21 (Linux 2.4.36) (91%)
- No exact OS matches for host (test conditions non-ideal) suggests that the test conditions might not have been ideal. This could be due to Network interference, Firewall configurations blocking, Customize kernel settings
- **Operating System:** The target appears to run a Linux-based system.
- **Exposed Services:** While many services are necessary for functionality, while exposing protocols like FTP increases risk. These should be reviewed for security best practices.
- **Firewall Protection:** Filtered ports indicate effective firewall rules, but open services still pose risks.
- **Discuss the ethical and legal considerations that security professionals should be aware of and adhere to during this process.**
 1. Security professionals must always obtain proper authorization before scanning any network, as unauthorized testing is illegal and unethical.
 2. They must respect privacy and only collect data relevant to the assessment, avoiding access to sensitive or personal information.
 3. Scans should be non-intrusive to avoid disrupting services or causing system crashes.
 4. Professionals must stay within the agreed scope and do not exploit vulnerabilities without permission.
 5. Findings should be reported responsibly to authorized persons only, ensuring confidentiality and integrity.

4. Scanning Beyond IDS and Firewall:

Command: `nmap -f --scan-delay 3s -sS --top-ports 200 162.241.216.11`

- **-f(fragmented):** Breaks packets into smaller pieces harder for simple firewalls/IDS to analyze.
- **--scan-delay 3s:** Send one packet after every 3 seconds
- **-sS:** SYN scan - sends SYN packets without completing the TCP handshake (stealthy and non-intrusive)
- **--top-ports 200:** Scans Top 200 ports
- **162.241.216.11:** Target domain IP

```

(kali@kali)-[~]
$ nmap -f --scan-delay 3s -sS --top-ports 200 162.241.216.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 11:44 EDT
Nmap scan report for 11.216.241.162.in-addr.arpa (162.241.216.11)
Host is up (0.28s latency).
Not shown: 180 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    open  smtp
26/tcp    open  rsftp
53/tcp    open  domain
80/tcp    open  http
110/tcp   open  pop3
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
143/tcp   open  imap
443/tcp   open  https
445/tcp   filtered microsoft-ds
465/tcp   open  smtps
587/tcp   open  submission
993/tcp   open  imaps
995/tcp   open  pop3s
1022/tcp  filtered exp2
1026/tcp  filtered LSA-or-nterm
3306/tcp  open  mysql
5432/tcp  open  postgresql

Nmap done: 1 IP address (1 host up) scanned in 716.97 seconds

```

Key Data Extracted:

- **Host is up:** The target is active and reachable.
- **Open Ports:** Several ports are open (e.g., FTP, SSH, SMTP, HTTP, HTTPS).
- **Filtered Ports:** Some ports (like msrpc, netbios-ssn, microsoft-ds) are marked as filtered, indicating firewall rules blocking them.
- **Slow Timing:** The scan took 716.97 seconds, which is very slow — likely to avoid detection by rate-based IDS.

Command: `nmap -f --scan-delay 10s -sS -p 135,139,445,1022,1026 -D RND:15 162.241.216.11`

- **-f(fragmented):** Breaks packets into smaller pieces harder for simple firewalls/IDS to analyze.
- **--scan-delay 10s:** Send one packet after every 10 seconds
- **-sS:** SYN scan - sends SYN packets without completing the TCP handshake (stealthy and non-intrusive)
- **-p 135,139,445,1022,1026:** Specific ports to scan
- **-D RND:15:** Adding random 15 decoy IPs to mask original IP

```

(kali@kali)-[~]
$ nmap -f --scan-delay 10s -sS -p 135,139,445,1022,1026 -D RND:15 162.241.216.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 12:04 EDT
Nmap scan report for 11.216.241.162.in-addr.arpa (162.241.216.11)
Host is up (0.30s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds
1022/tcp   filtered  exp2
1026/tcp   filtered  LSA-or-nterm

Nmap done: 1 IP address (1 host up) scanned in 121.68 seconds

```

Key Data Extracted:

- Host is up: The target remains active.
- Filtered Ports: All scanned ports (135/tcp, 139/tcp, 445/tcp, 1022/tcp, 1026/tcp) are still marked as filtered.
- Decoy IPs: Added 15 random decoy IPs to mask your real IP address.
- Slow Timing: The scan took 121.68 seconds, slower than a normal scan.
- Despite using decoy IPs and increasing the scan delay to 10 seconds, the targeted ports remained filtered, suggesting that the firewall continues to block the traffic.
- **Vulnerabilities and potential risks that may have been missed by these security measures.**
 - I. IDS May Be Weak or Misconfigured
 1. The slow scan (--scan-delay 10s) completed without triggering any alerts.
 2. This means the IDS does not detect low and slow scans, which real attackers often use.
 3. Risk: An attacker could slowly examine the network over hours/days without being noticed.
 - II. Open Services Still Exposed
 1. Critical services like FTP (21), SSH (22), SMTP (25), HTTP (80), HTTPS (443) remain open.
 2. Example Risk: FTP is unencrypted - could leak credentials if intercepted.

5. Network Scan Using Wireshark:

Step 1: Open Wireshark and Select Interface

- Launch Wireshark.
- Choose the active network interface:
 - Wi-Fi (wlan0)
 - Ethernet (eth0)
- Click capturing packet by click on shark icon

Step 2: Generate traffic with port scan

- In Terminal, Run **nmap -sS --top-ports 20 162.241.216.11**

- This sends SYN packet to capture ports

Step 3: Apply a capture Filter

- In Wireshark use **ip.addr == 162.241.216.11 && tcp.flags.syn**

- This capture traffic from and to Target.

Step 4: Analyze the traffic

Open Ports:

192.168.1.10 (My IP) to 162.241.216.11 (Target IP): [SYN]

162.241.216.11 (Target IP) to 192.168.1.10 (My IP): [SYN, ACK]

Closed Ports:

192.168.1.10 (My IP) to 162.241.216.11 (Target IP): [SYN]

162.241.216.11 (Target IP) to 192.168.1.10 (My IP): [RST, ACK]

Filtered Ports (Blocked by Firewall):

192.168.1.10 (My IP) to 162.241.216.11 (Target IP): [SYN]

No Response

ip.addr == 162.241.216.11 && tcp.flags.syn						
No.	Time	Source	Destination	Protocol	Length	Info
7	6.169827022	192.168.1.10	162.241.216.11	TCP	58	47869 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
8	6.170414431	192.168.1.10	162.241.216.11	TCP	54	47869 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
11	6.432326284	162.241.216.11	192.168.1.10	TCP	60	443 → 47869 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
12	6.432373217	192.168.1.10	162.241.216.11	TCP	54	47869 → 443 [RST] Seq=1 Win=0 Len=0
15	6.477043538	192.168.1.10	162.241.216.11	TCP	58	48125 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
16	6.477312794	192.168.1.10	162.241.216.11	TCP	58	48125 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
17	6.477473790	192.168.1.10	162.241.216.11	TCP	58	48125 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
18	6.477639047	192.168.1.10	162.241.216.11	TCP	58	48125 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
19	6.477812533	192.168.1.10	162.241.216.11	TCP	58	48125 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
20	6.477942204	192.168.1.10	162.241.216.11	TCP	58	48125 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
21	6.478098328	192.168.1.10	162.241.216.11	TCP	58	48125 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
22	6.478309395	192.168.1.10	162.241.216.11	TCP	58	48125 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
23	6.478488063	192.168.1.10	162.241.216.11	TCP	58	48125 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
24	6.478613965	192.168.1.10	162.241.216.11	TCP	58	48125 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
25	6.732938987	162.241.216.11	192.168.1.10	TCP	60	25 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
26	6.732939900	162.241.216.11	192.168.1.10	TCP	60	5900 → 48125 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
27	6.732940040	162.241.216.11	192.168.1.10	TCP	60	80 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
28	6.732940160	162.241.216.11	192.168.1.10	TCP	60	143 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
29	6.733222277	192.168.1.10	162.241.216.11	TCP	54	48125 → 25 [RST] Seq=1 Win=0 Len=0
30	6.733528051	192.168.1.10	162.241.216.11	TCP	54	48125 → 80 [RST] Seq=1 Win=0 Len=0
31	6.733839778	192.168.1.10	162.241.216.11	TCP	54	48125 → 143 [RST] Seq=1 Win=0 Len=0
32	6.735171132	162.241.216.11	192.168.1.10	TCP	60	8080 → 48125 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
33	6.737711506	162.241.216.11	192.168.1.10	TCP	60	111 → 48125 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
34	6.740695083	192.168.1.10	162.241.216.11	TCP	58	48125 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
35	6.741142998	192.168.1.10	162.241.216.11	TCP	58	48125 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
36	6.741309377	192.168.1.10	162.241.216.11	TCP	58	48125 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
37	6.741498641	192.168.1.10	162.241.216.11	TCP	58	48125 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
38	6.741680567	192.168.1.10	162.241.216.11	TCP	58	48125 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
39	6.741808203	192.168.1.10	162.241.216.11	TCP	58	48125 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
40	6.741942094	192.168.1.10	162.241.216.11	TCP	58	48125 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
41	6.742056619	192.168.1.10	162.241.216.11	TCP	58	48125 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
42	6.742200604	192.168.1.10	162.241.216.11	TCP	58	48125 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
43	6.742387212	192.168.1.10	162.241.216.11	TCP	58	48125 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
44	6.746120157	162.241.216.11	192.168.1.10	TCP	60	22 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
45	6.746122543	162.241.216.11	192.168.1.10	TCP	60	110 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
46	6.746223505	192.168.1.10	162.241.216.11	TCP	54	48125 → 22 [RST] Seq=1 Win=0 Len=0
47	6.746377454	192.168.1.10	162.241.216.11	TCP	54	48125 → 110 [RST] Seq=1 Win=0 Len=0
48	6.990751632	162.241.216.11	192.168.1.10	TCP	60	3306 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
49	6.991026251	192.168.1.10	162.241.216.11	TCP	54	48125 → 3306 [RST] Seq=1 Win=0 Len=0
50	6.992946832	162.241.216.11	192.168.1.10	TCP	60	993 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
51	6.993095438	192.168.1.10	162.241.216.11	TCP	54	48125 → 993 [RST] Seq=1 Win=0 Len=0
52	7.005868990	162.241.216.11	192.168.1.10	TCP	60	1723 → 48125 [RST, ACK] Seq=1 Ack=1 Win=32120 Len=0 MSS=1440
53	7.006601477	162.241.216.11	192.168.1.10	TCP	60	443 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
54	7.006691142	192.168.1.10	162.241.216.11	TCP	54	48125 → 443 [RST] Seq=1 Win=0 Len=0
55	7.008251685	162.241.216.11	192.168.1.10	TCP	60	53 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
56	7.008412241	192.168.1.10	162.241.216.11	TCP	54	48125 → 53 [RST] Seq=1 Win=0 Len=0
57	7.010198698	162.241.216.11	192.168.1.10	TCP	60	23 → 48125 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
58	7.015611376	162.241.216.11	192.168.1.10	TCP	60	3389 → 48125 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
59	7.015611857	162.241.216.11	192.168.1.10	TCP	60	995 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1360
60	7.015611968	162.241.216.11	192.168.1.10	TCP	60	21 → 48125 [SYN, ACK] Seq=0 Ack=1 Win=32120 Len=0 MSS=1360
61	7.015716869	192.168.1.10	162.241.216.11	TCP	54	48125 → 995 [RST] Seq=1 Win=0 Len=0
62	7.015873896	192.168.1.10	162.241.216.11	TCP	54	48125 → 21 [RST] Seq=1 Win=0 Len=0
63	7.868234899	192.168.1.10	162.241.216.11	TCP	58	48127 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
64	7.868498782	192.168.1.10	162.241.216.11	TCP	58	48127 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
65	7.868613246	192.168.1.10	162.241.216.11	TCP	58	48127 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Key TCP Flags to Look For:

• SYN (Synchronize):

Sent to a server establish a connection.

• SYN-ACK (Synchronize-Acknowledge):

Sent by the server if the port is open, acknowledging the SYN packet.

• RST-ACK (Reset-Acknowledge):

Sent by the server if the port is closed, resetting the connection attempt.

- **No Response:**

Indicates the port is filtered (likely blocked by a firewall).

- **Open Ports Observed:**

Ports that responded with a SYN-ACK are considered open because they acknowledged the connection attempt.

Destination Port	Port Name	Response	Conclusion
80	HTTP	SYN, ACK	Open
443	HTTPS	SYN, ACK	Open

- **Closed Port Observed:**

Ports that responded with an RST-ACK are considered closed because they actively reject the connection attempt.

Destination Port	Port Name	Response	Conclusion
22	SSH	RST, ACK	Closed
25	SMTP	RST, ACK	Closed

- **Filtered Ports:**

Ports that showed no response (i.e., no ACK or RST) are considered filtered because the firewall is likely blocking the packets.

Destination Port	Port Name	Response	Conclusion
135	msrpc	No response	Filtered

6. Generating a Comprehensive Report:

A network security assessment was carried out on **www.certifiedhacker.com**, a publicly available website meant for learning and ethical testing. The goal was to find out which devices are active, what ports are open, what operating system is running, and if there are any potential security weaknesses - all while following proper ethical and legal practices.

Using tools like Nmap, ping, and Wireshark, we found several services running, including web, email, and remote access. While the firewall did a good job blocking unnecessary ports, some open services could still be risky if not properly secured. Importantly, no harmful or aggressive tests were performed - everything was done safely and responsibly.

Key Findings:

- Host is active (confirmed via ping)
- Multiple services open: HTTP (80), HTTPS (443), FTP (21), SSH (22), SMTP (25), and email ports
- OS likely Linux-based (kernel 2.6-3.10)
- Firewall blocks ports like 135, 139, 445 - good protection
- IDS did not detect slow or decoy scans - weak detection capability

Vulnerabilities & Risks:

- FTP is unencrypted - could leak credentials
- SSH and SMTP exposed - potential for brute-force attacks

- Slow scans went undetected - attacker could examine silently

Recommendations:

- Replace FTP with SFTP or disable it
- Harden SSH (use key-based login, change port)
- Update all services (Apache, OpenSSH, Exim)
- Improve IDS rules to detect slow and fragmented scans
- Regularly scan and monitor logs

Conclusion:

The security assessment of the network was completed using approved and ethical methods. Tools like ping, Nmap, and Wireshark were used to safely check the target system.

We found that the host is active and running several services, such as a web server (port 80 and 443), email services, and SSH access. Some ports are open, which is necessary for the system to work, but they can be risky if not protected well.

The firewall is doing a good job by blocking unnecessary ports like 135 and 139. However, the Intrusion Detection System (IDS) did not detect slow or hidden scans, which means an attacker could try to sneak in without being noticed.

No harmful actions were taken - all steps followed ethical and legal rules.

Overall, the network has basic protection, but improvements are needed. By closing unused services, updating software, and making the IDS smarter, the system can become much more secure.